PAPER ID-311705

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM III) THEORY EXAMINATION 2025-26
ENGINEERING MECHANICS

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 07 = 14

Q no.	Question	CO
a.	What is a coplanar concurrent force system?	1
b.	Define mass moment of inertia.	2
c.	State the relation between load, shear force and bending moment.	3
d.	What are the types of coordinate systems used in curvilinear motion?	4
e.	Write the equations of motion for uniformly accelerated rectilinear motion.	4
f.	What is meant by angular impulse?	5
g.	State the work–energy principle for rigid bodies.	5

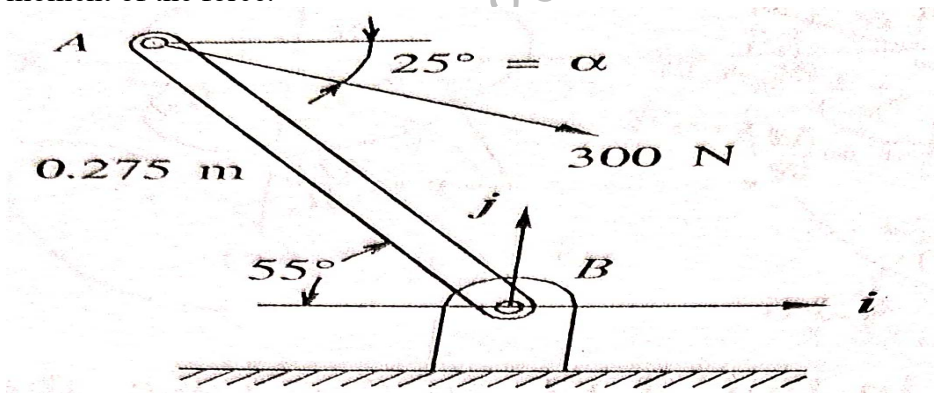
SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

Q no.	Question	CO
a.	Find the resultant of three concurrent forces of 40 N, 50 N and 60 N acting at angles of 30°, 60° and 120° respectively.	1
b.	Define area moment of inertia and derive the moment of inertia of a rectangular section about its centroidal axis from first principles.	2
c.	A simply supported beam of length 6 m carries a UDL of 4 kN/m and a point load of 12 kN at 2 m from left support.Draw SFD and BMD.	3
d.	Explain polar coordinates and derive expressions for velocity and acceleration.	4
e.	Derive the work–energy equation for a rigid body in plane motion.	5

SECTION C

3. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	A 300 N force is applied to the control rod as shown in figure. Knowing that the length of the rod is 275 mm and α = 25º, determine moment of the force. 	1
b.	Explain the working principle of a screw jack with neat sketch.	1

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4. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Find the moment of inertia of a circular plate of diameter 200 mm about its centroidal axis.	2
b.	State and prove the perpendicular axis theorem.	2

5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Explain zero force members and state the rules for identifying zero force members.	3
b.	Draw SFD and BMD for a cantilever beam carrying UDL over its entire length.	3

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Two smooth balls of masses 2 kg and 4 kg move towards each other with velocities 6 m/s and 2 m/s respectively. If the coefficient of restitution is 0.75, find the velocities after impact.	4
b.	A particle moves along a curve such that r = 60t2-20t2 and θ = 2t2 is in mm and θ in radians. Determine the velocity and acceleration of the particle when t = 1 second.	4

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Explain the application of virtual work method to connected bodies.	5
b.	Two equal balls meet centrally with speed 2 m/s and -3 m/s. What will be their final speeds after impact? The coefficient of restitution is 0.8.	5

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